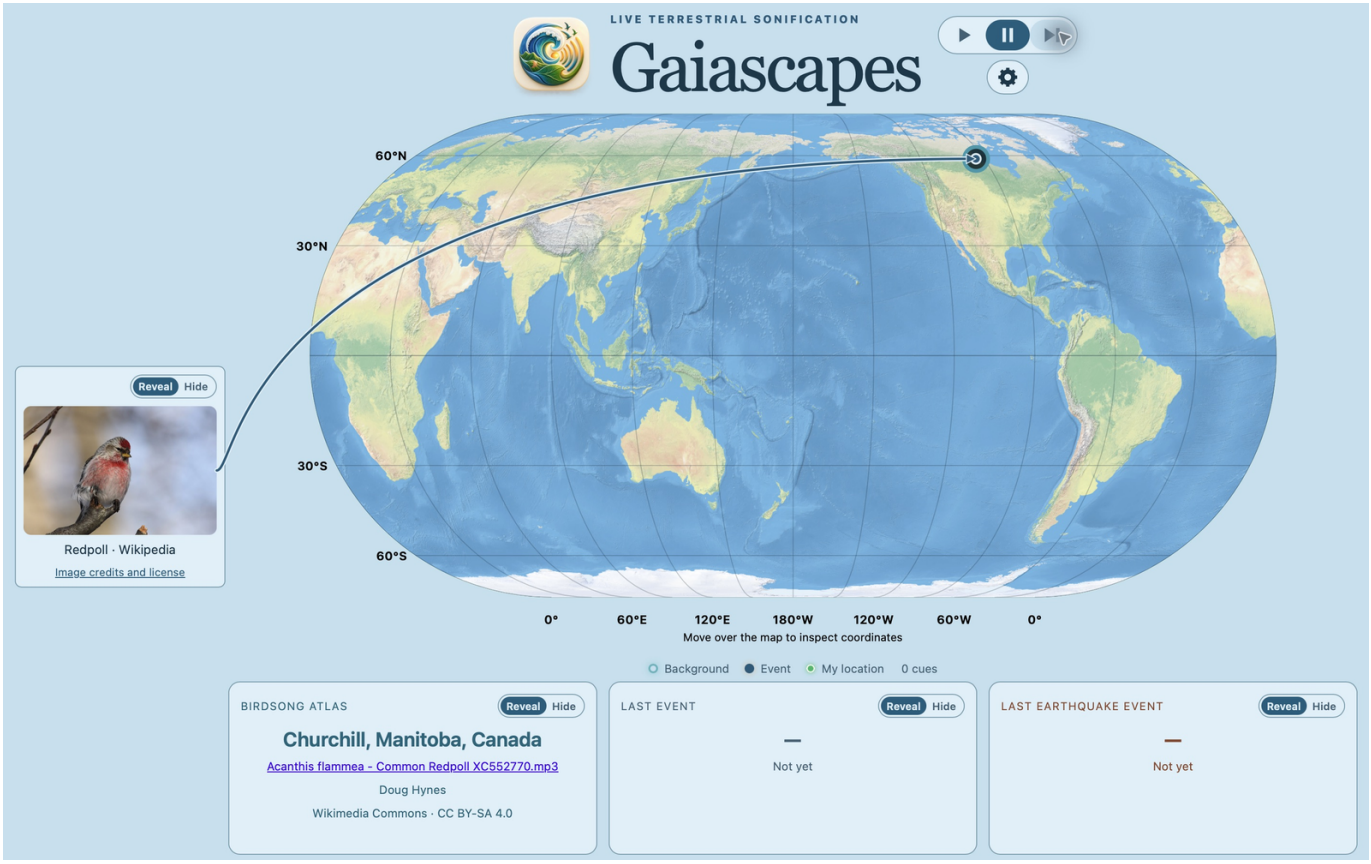
Gaiascapes User Guide



Pacific-centered map with a staged Redpoll image, its recording-location pointer, Play/Pause/Next controls, and status tiles

This guide explains the controls in Gaiascapes. Installation and system setup are covered separately in the README.

Peace Hill Studios hope you find the app fun and educational with soothing background sounds.

All screenshots were refreshed from this version on 29 October 2026 using a separate temporary installation. Enabled sources and selected sounds are examples, not the factory defaults. The mobile images show Safari's 390 × 844 responsive viewport.

Map View

The map is always visible in the main window, with the themed animation behind it.

The legend distinguishes background locations, event locations, and **My location**. Moving the pointer over the map displays coordinates. The system-location marker, when available, shows the host's approximate location derived from its public IP address. Background locations come from **Sound locations**; the host location can seed location 1 for Xeno-canto Birdsong and Storm Outlook at startup (see Location privacy).

The main view is the map, with the animated themed background behind it. Use the **Play / Pause / Next** control strip beside the title. Playback status and errors appear beside the controls. Open Settings with the **gear (⚙)** in its themed

pill, centered beneath the playback strip. Settings contains **Sound sources**, **Sound Choices**, **Sound locations**, **Display**, and finally **Event History**. The version appears above the Settings heading.

Start a soundscape

1. Open Gaiascapes and select the **gear (⚙)** to open Settings.
2. In **Sound sources**, enable the source you want. **Birdsong → Wikimedia Commons** works without an API key. Sources that need credentials show their fields when enabled.
3. In **Sound Choices**, use the arrows under **Background** to choose a sound, and set its volume. Choose event sounds independently, or select **None** for quiet event slots.
4. Select **Save settings** and wait for **Settings saved.** Close Settings with **×**.
5. Press **Play**. On another device, also enable **Settings → Sound Choices → Listen on this device**. Press **Pause** to stop the shared soundscape; **Mute this device** affects only that listening browser.

For synthesized backgrounds or event voices, the host's SuperCollider renderer must be running. Recorded animal sounds, the map, settings, and event history can operate without it.

Playback controls

Playback uses continuous sound: new event voices play over a rotating background. **Play** starts or resumes the soundscape; **Pause** pauses it while enabled feeds continue collecting observations. **Next** stages the next animal image and pauses without playing its recording. Capture/replay remains available through the API; the UI has no mode selector or replay controls.

Ocean Swells and Storm Outlook normally advance to another background location approximately every **23 seconds**. Wildlife recordings use their audio length instead. Network searches and downloads can take longer. Wildlife recordings are archived audio, even while the app is in Continuous mode.

For **Birdsong, Frog Calls, Whale Song, and Dolphin Calls**, each location visit plays one complete recording once. Short recordings do not loop, and long recordings are not cut off after 23 seconds. The player requests the next location during the final **three seconds**, or final **10%** of a short clip, and fades between recordings while the outgoing clip finishes. A first-time download can leave a gap between clips. Keep a browser opened locally on the host, or the host's desktop player, open for the recording rotation to advance. A remote listening browser does not advance that rotation. **Preview** remains an eight-second sample; **Pause** or changing the background can interrupt a recording.

Wikimedia Commons Birdsong, Xeno-canto Birdsong, Frog Calls, Whale Song, and Dolphin Calls each have **19 default locations**. Mammal collections use their smaller, fixed site lists. The first pass plays the first available recording at each location. The next pass plays the second recording at each location, and later passes continue through each location's list before wrapping. A location with one verified recording repeats it on each pass. Marine playback follows the included subset when you exclude sites.

Status tiles

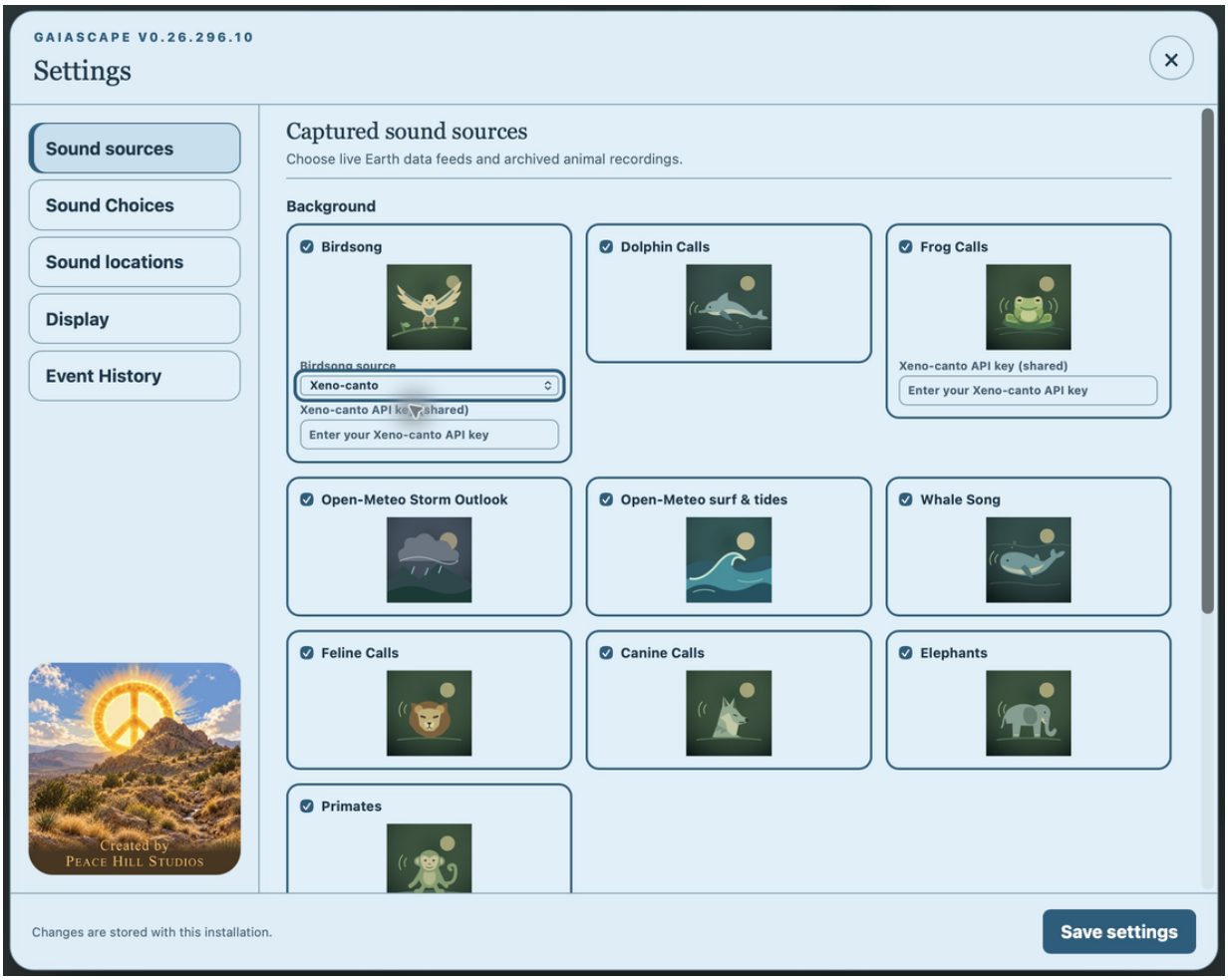
Three separate status tiles appear below the map. On desktop they sit side by side along the bottom; on mobile they stack vertically:

Card	Meaning
Background	Uses the selected sound's name, such as Birdsong Atlas or Frog Calls. Shows the most recent matching location and its measurements or recording attribution. It also reports recording searches and lookup failures.
Last Event	The most recently presented event type and time.
Last Earthquake Event	The latest available earthquake location and details, including captured observations.

A dash or **Not yet** means there is no matching event to display yet. A previous event can remain visible after playback stops; it is not evidence that sound is still playing.

Sound sources

Open **Settings** → **Sound sources**. Background sources appear first: Birdsong through Whale Song, followed by the four mammal groups and Announce Recording. Event sound sources appear below them. Click a checkbox or its artwork to enable or disable that source. Hover over a title, or focus its checkbox, for its explanation.



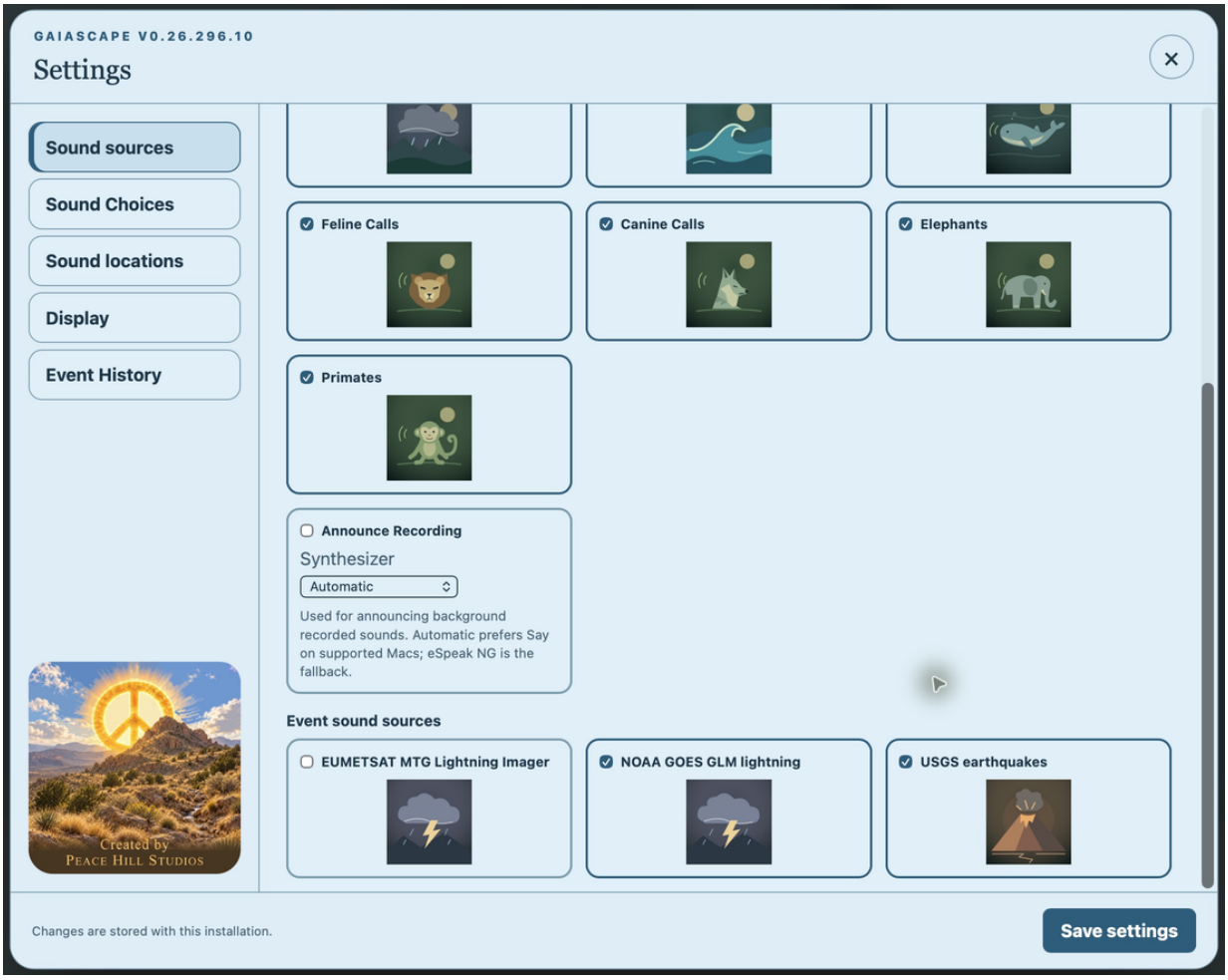
Sound sources with Xeno-canto selected for Birdsong and empty shared API-key fields for Birdsong and Frog Calls

Sound sources determines which providers are available. **Sound Choices** determines which of their sounds play. Enabling several background sources does not mix them together: there is one Background sound selection. Save source changes before using Preview.

Feline Calls, Canine Calls, Elephants, and Primates each have an animal tile with an enable checkbox in Background sound sources and an entry in the Sound Choices Background selector. They are disabled initially. The bundled starter pack contains nine recordings: lion and tiger; dingo, fox, and coyote; one elephant soundscape; bonnet macaque and two howler monkey recordings. Enable the desired source and select it as Background to play it. No key or recording download is needed.

Proof-of-concept limitation: the current elephant clip has been reported as dominated by other wildlife rather than a usable elephant call. It needs replacement and is not an approved elephant example; the source remains disabled by default.

These groups use only fixed recording sites, with no user-selectable locations. Sequential/Random, Preview, Next, images, and announcements use those sites. Zoo and other recording context appears in the status information. Broader animal labels are used where species identification is unresolved. See the recording credits and curation notes. **Announce Recording** is the last background source tile.



Mammal source tiles followed by Announce Recording and its synthesizer selector

Background sources

Source tile	What it supplies	Credentials and additional settings
Birdsong	Archived bird recordings. The **Birdsong source** selector chooses Wikimedia Commons or Xeno-canto.	**Wikimedia Commons:** no key; uses 19 curated locations. **Xeno-canto:** shared API key; uses 19 editable regions with searches within 100 km of each center.
Dolphin Calls	Dolphin recordings from NOAA and other verified archives at 19 locations.	No key. Select included sites in Sound locations.
Frog Calls	Xeno-canto frog and toad recordings near the chosen region centers.	The same Xeno-canto API key as Birdsong. Its 19 region centers are edited independently of Birdsong.
Open-Meteo Storm Outlook	Forecast conditions used to generate a storm background.	No key. Uses the 19 Storm Outlook locations. This is forecast convection, not observed lightning.
Open-Meteo surf & tides	Modeled ocean swells and tide turns.	No key. Uses the 19 Ocean Swells locations. Supplies both the Ocean Swells background and events for Tidal Tone.
Whale Song	NOAA whale songs and calls from several species at 19 locations.	No key. Select included sites in Sound locations.
Feline Calls	Bundled lion and tiger recordings at fixed sites.	No key or download; no editable location catalog.
Canine Calls	Bundled dingo, fox, and coyote recordings at fixed sites.	No key or download; no editable location catalog.
Elephants	One bundled soundscape at a fixed site; see the proof-of-concept limitation above.	No key or download; disabled by default.
Primates	Bundled bonnet macaque and howler monkey recordings at fixed sites.	No key or download; no editable location catalog.

NOAA marine recordings come from the NCEI passive acoustic archive. Gaiascapes caches selected recordings locally and retains their source attribution. A first use can require a download; subsequent uses can reuse the cache.

Event sound sources

Source tile	What it supplies	Credentials
EUMETSAT MTG Lightning Imager	Observed lightning flashes covering Europe and Africa, used by Thunder.	**Consumer Key** and **Consumer Secret**, obtained through EUMETSAT.
NOAA GOES GLM lightning	Observed total-lightning flashes from GOES-East and GOES-West, used by Thunder.	No API key.
USGS earthquakes	Earthquake observations, used by Earthquake and Seismic Tone.	No API key.

Credentials are required only for **Birdsong using Xeno-canto**, **Frog Calls**, and **EUMETSAT MTG Lightning Imager**. There is no credential field for the other sources.

Obtaining credentials

Xeno-canto: one key for Birdsong and Frog Calls

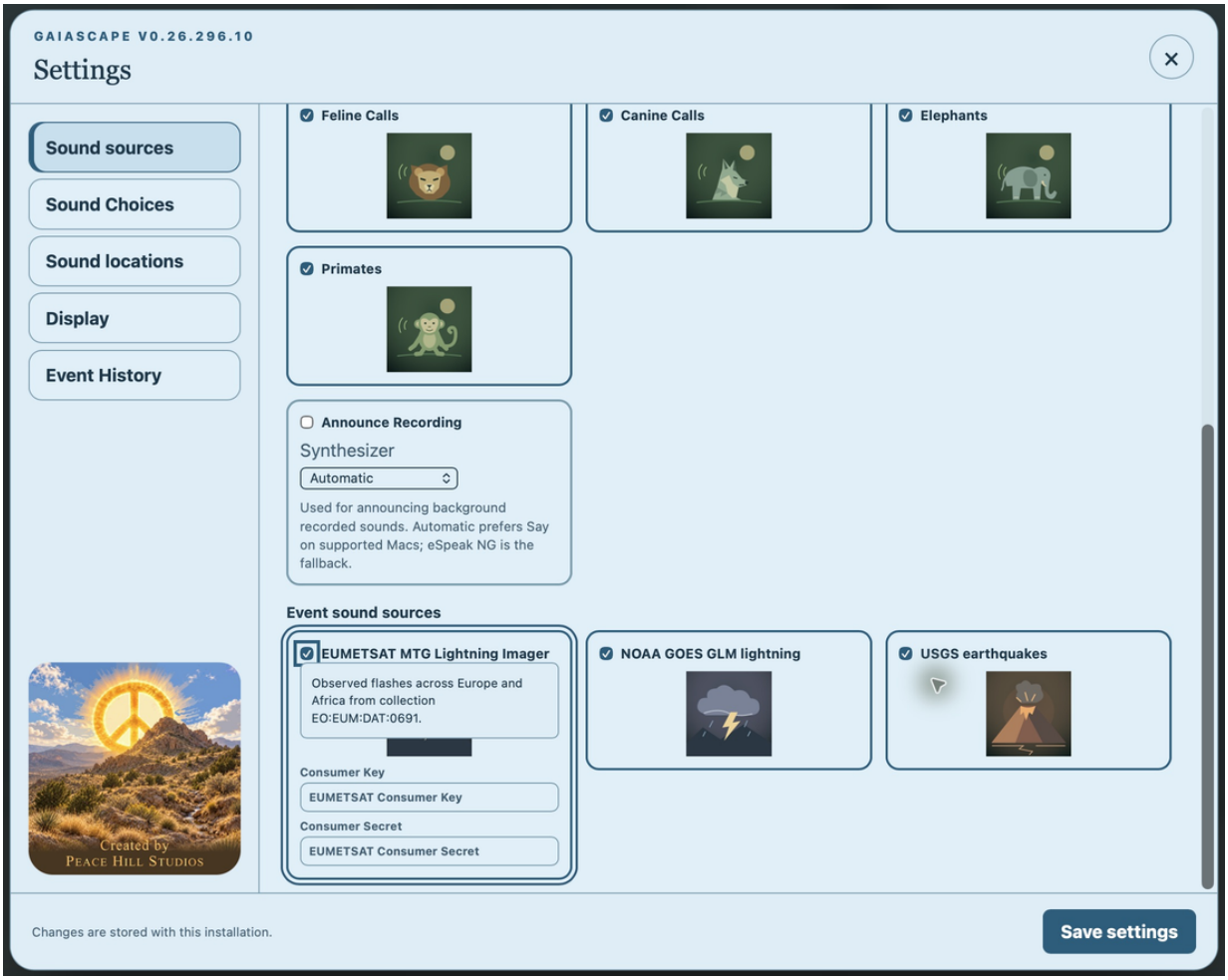
1. Open Xeno-canto in your browser and sign in, or register an account. Complete any email-verification step requested by the site.
2. Open Your account and locate your API key. The official API page provides the provider's current API guidance. These pages may require sign-in or a browser verification check.
3. In Gaiascapes, open **Settings** → **Sound sources**.
4. Either enable **Birdsong**, choose **Xeno-canto** in **Birdsong source**, and enter the key; or enable **Frog Calls** and enter it in that tile's **Xeno-canto API key (shared)** field.
5. Click **Save settings**. One saved key serves both sources. Replacing it in either field changes the shared key.

Leave the key field blank when you want to keep the saved key. After a successful save, the field clears and its placeholder indicates that a key is saved. Birdsong using Wikimedia Commons does not need this key.

The account-page link is also documented by the rOpenSci suwo reference. Xeno-canto blocked automated inspection of its account pages during preparation of this guide, so exact account-page wording could not be checked. No user credentials or recording API requests were used to prepare these instructions.

EUMETSAT: Consumer Key and Consumer Secret

Scroll down in Sound sources if the credential fields are below the visible area.



Sound sources scrolled to show EUMETSAT Consumer Key and Consumer Secret

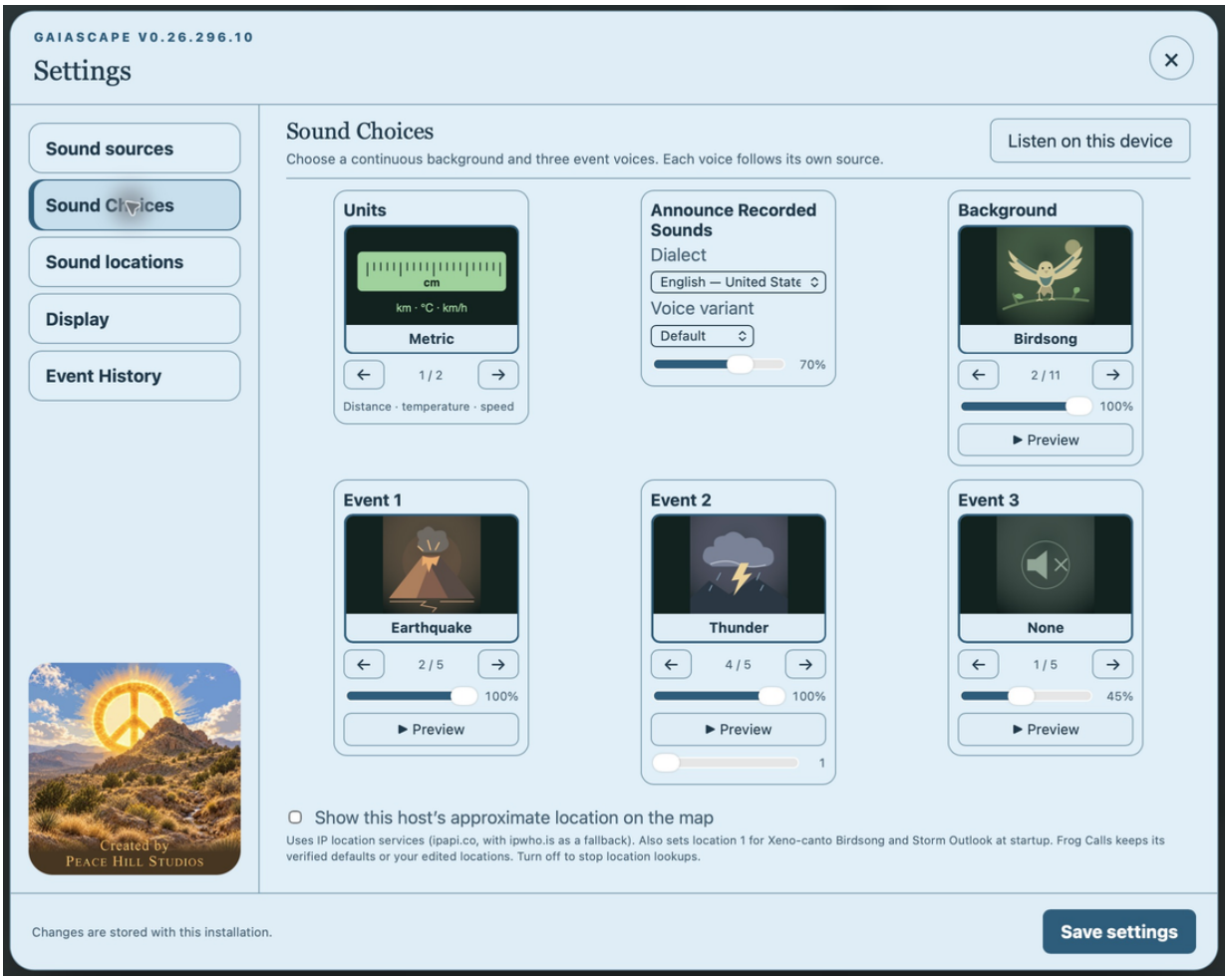
1. Open the EUMETSAT User Portal sign-in page. Choose **Register - Create new account** if needed, then complete registration and sign in.
2. Open the EUMETSAT Data Store.
3. Open the menu under your username and select **API Key**. You can also open API key management directly after signing in.
4. Under **User Credentials**, reveal and copy the **Consumer Key** and **Consumer Secret**. Use these two values, rather than the temporary access token.
5. In Gaiascapes, enable **Settings** → **Sound sources** → **EUMETSAT MTG Lightning Imager**, paste the two values into the matching fields, and click **Save settings**.

EUMETSAT illustrates these steps in its Introductory Data Store user guide. If access is denied after authentication, check the account's applicable data licences using the registration and licensing guide.

Blank credential fields preserve their saved values. The **Saved - enter only to replace** placeholder indicates that credentials are stored. Gaiascapes handles token acquisition; you do not need to paste a generated token into the app.

Sound Choices

Open **Settings** → **Sound Choices**. The listening button is on the right of the heading. The first row contains **Units**, **Announce Recorded Sounds**, and **Background**. Below it are three independent event tiles: **Event 1**, **Event 2**, and **Event 3**. The location-privacy checkbox is beneath the sound tiles. These controls stack on narrow screens.



Sound Choices showing listening controls, Units, announcement dialect and voice variant, Birdsong, and three event voices

Listening from another device

Open http://<Gaiascapes-host-IP>:8769 from a device on the same LAN, then tap **Listen on this device** in **Settings** → **Sound Choices**, at the right of the Sound Choices heading. **Mute this device** stops playback in that browser without stopping the performance, the host speakers, or other listeners. New remote pages start silent; tap Listen again after reloading. **Play** and **Pause** control the shared soundscape on the host.

The host must be running and reachable from the listening device. Use the host's LAN address, not localhost or 127.0.0.1, on another device. The supplied launchers accept LAN connections on port **8769**; if the page will not open, check the host address, firewall, and whether the Wi-Fi network isolates clients. See Ports and LAN access for host binding details. Listening uses the same web connection; no separate audio port needs to be opened on the LAN.

The browser combines live SuperCollider audio with the animal recordings served by Gaiascapes. Only Gaiascapes audio is included. Update and restart both the Gaiascapes server and its SuperCollider receiver before using this feature. SuperCollider must run on the same host as the server. No loopback audio driver or microphone permission

is required. With SuperCollider disabled, Listen plays animal recordings only and explains that limitation beside the button.

Keep the page open while listening. If the browser suspends audio or the connection drops, the message beside Listen explains how to reconnect. Remote playback has a short buffer and may lag the host speakers. Each browser controls its own playback; listening locally as well as through host speakers can produce an echo. Remote listeners do not advance the host's recording rotation.

Listening and muting take effect immediately and do not require **Save settings**. Closing Settings leaves device playback active. The relay supports up to **16 simultaneous listeners**; if it is full, mute an unused listener and retry.

Selecting a sound

Use the left and right arrows beneath a tile’s artwork to move through choices. Each sound list starts with **None**, followed by the remaining choices in A-Z order. The position indicator shows the selected item’s position and the number of choices. You can also use Left/Right while a sound card has keyboard focus; Home goes to the first choice and End to the last.

Each role has its own **volume slider**, from **0-100%**. Zero mutes that role. Select **None** to leave the role unassigned. Adjusting one role’s volume does not change the others. Every slider uses the same squared response curve: 10% gives -40 dB, 50% gives -12 dB, and 90% gives about -1.8 dB relative to that sound at 100%. Save updates sounds already playing as well as subsequent cues; Preview uses the displayed level. Synthesized voices receive gain independently of their musical velocity, including an exact zero target for mute. Different instruments and recordings still have different natural loudness.

Recorded birds, frogs, whales, and dolphins are automatically volume-normalized before playback, including previews. Louder recordings are reduced toward a quiet background level (-30 dBFS gated RMS); quiet recordings are never boosted. Peak protection also controls transitions without adding gain. The Background volume slider controls the listening level directly through Web Audio with a squared gain curve: 10% is -40 dB relative to full volume, 50% is -12 dB, and 100% is the full normalized level. Saving applies its new level immediately to the recording already playing, including during transitions. At zero volume, recordings continue advancing silently; saving a higher volume restores sound without waiting for the recording to finish. The browser analyzes each recording on first use, which can briefly delay playback, and remembers its level for the current page session. Original recordings are unchanged. This uses gated RMS loudness estimation, so recordings with very different frequency content may still sound somewhat different in loudness. A browser that cannot analyze a recording shows a playback error instead of playing it at an uncontrolled level.

Preview tries the displayed sound at the displayed volume. On another device, enable **Listen on this device** first to hear previews there. Synthesized previews also play through the host renderer. Recorded wildlife previews last approximately eight seconds. Preview is disabled for None. Save source, credential, or region edits before previewing them; moving to another sound and previewing it does not itself save that sound as your normal selection.

Units

Choose **Imperial** or **Metric** to control displayed measurements, such as distance, swell height, wind speed, and temperature where shown. The Units tile is a display preference, not an audio layer. Latitude and longitude remain geographic coordinates in either setting.

Recorded sound announcements

In **Sound Sources**, enable **Announce Recording** below the Background sources and choose its **Synthesizer**. Automatic prefers Say on supported Macs; Say also falls back to eSpeak NG when the chosen native voice is unavailable. Select eSpeak NG to always use that engine. In **Sound Choices** → **Announce Recorded Sounds**, choose a **Dialect** and **Voice variant** (Default, Male, or Female), set the volume at the bottom, and save settings. Each newly played animal recording announces its species and named location before the animal audio starts. The previous animal sound stops before speech, so speech and animal audio do not overlap. Disabled announcements retain the usual recording transitions. If speech fails, its error is shown and the animal recording still plays. Coordinate-only locations are skipped. Dialects change pronunciation; names use the original recording metadata and are not translated. The default is disabled, US English, 70% volume.

Announcements use Say or eSpeak NG on the host; see the installation instructions. No speech service or network API is required. SuperCollider is needed only for Say. Muting this device, stopping playback, or moving to another recording cancels its current announcement. The tile reports a missing engine, missing voice, or synthesis failure while recorded sound playback remains available.

On macOS, Say starts a short-lived SuperCollider language (sclang) worker for each uncached announcement. A second SuperCollider Dock icon may appear briefly while it generates speech; it is not loading the animal recording. Choosing **eSpeak NG** in **Sound sources** → **Announce Recording** → **Synthesizer** avoids that extra process. Repeated announcements may use the in-memory speech cache.

Background choices

Choice	Sound and dependency
None	No continuous background. Event voices can still play.
Birdsong	Plays recordings from the provider chosen in the Birdsong source tile.
Canine Calls	Plays bundled canine recordings; enable Canine Calls in Sound sources.
Dolphin Calls	Plays curated dolphin recordings from the included sites.
Elephants	Plays the bundled elephant soundscape; enable Elephants and review the limitation above.
Feline Calls	Plays bundled lion and tiger recordings; enable Feline Calls.
Frog Calls	Plays Xeno-canto frog recordings from the configured regions.
Ocean Swells	Generates a background from modeled swell conditions; requires Open-Meteo surf & tides.
Primates	Plays bundled primate recordings; enable Primates.
Storm Outlook	Generates a background from forecast storm conditions; requires Open-Meteo Storm Outlook.
Whale Song	Plays curated whale songs and calls from the included sites.

Animal recordings, including the four mammal groups, play through the browser or desktop web view. Ocean Swells, Storm Outlook, and the event voices use the configured SuperCollider renderer. A working recording preview therefore does not by itself confirm that synthesized sounds are available.

Event 1, Event 2, and Event 3

Each slot offers the same choices. Slot numbers do not restrict which source it follows.

Choice	Trigger
None	Leaves this event slot silent.
Earthquake	USGS earthquake observations, rendered as a low-frequency earthquake sound.
Seismic Tone	USGS earthquake observations, rendered as a tonal voice.
Thunder	Observed lightning from enabled NOAA GOES GLM and/or EUMETSAT MTG sources. Storm Outlook forecasts do not trigger it.
Tidal Tone	Modeled tide-turn events from Open-Meteo surf & tides.

Selecting the same sound in multiple event slots creates multiple voices for matching events, each with its own volume. It does not select different geographic regions for those slots.

An older saved configuration may show **Retired sound**. It remains available as the current selection until you choose a replacement; it is not a new sound to configure.

Lightning sample rate

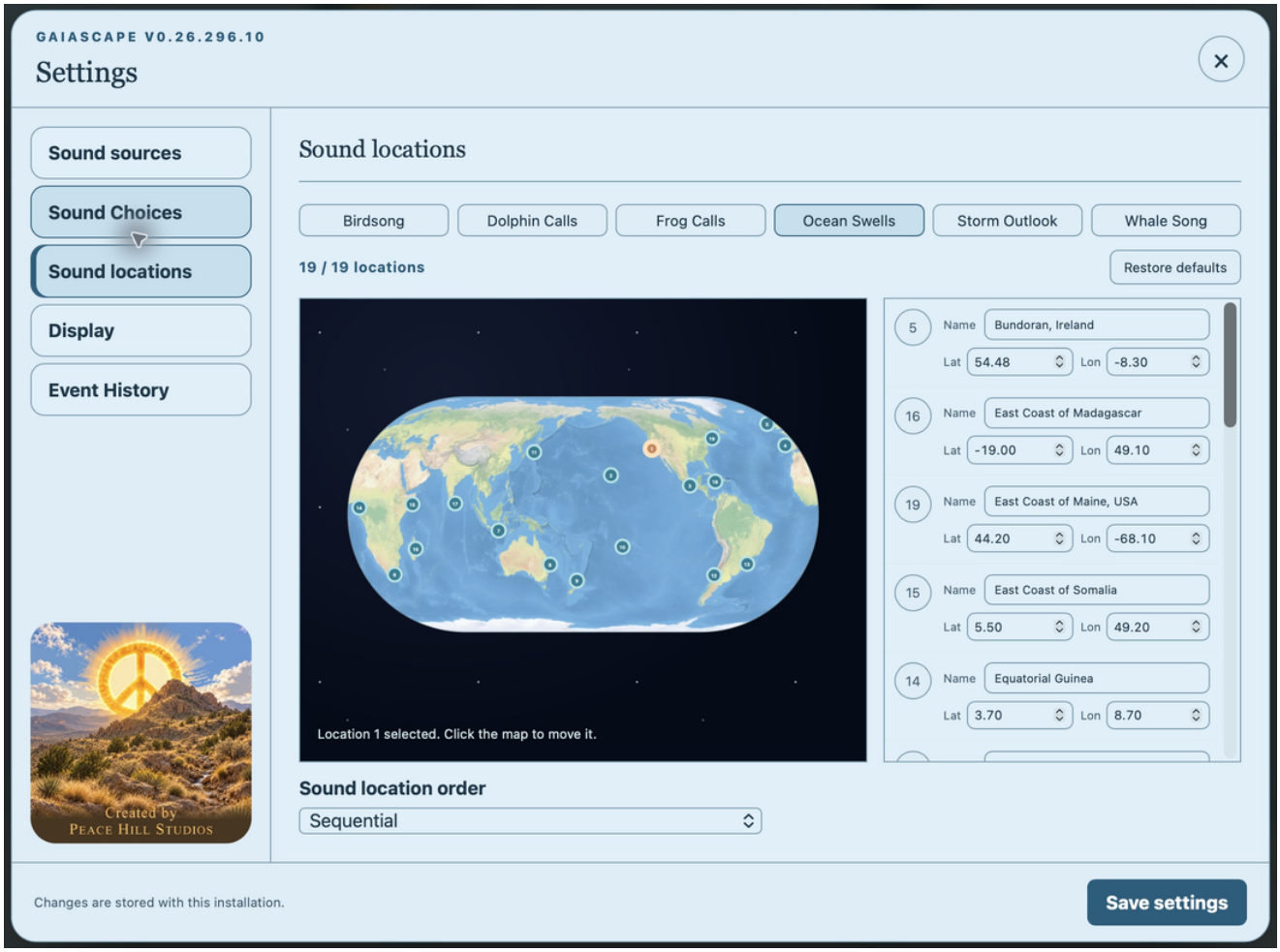
A **lightning sample rate** slider appears under event slots assigned to Thunder. Its range is **1-11**. A value of 1 uses every available flash selected for sonification; larger values thin the sound stream to approximately every Nth flash. For example, 5 is roughly one in five. This reduces sound density, rather than changing audio quality or the provider's polling interval.

The setting is shared: changing it under one Thunder slot updates the other visible sample-rate sliders. Each event slot's volume remains independent.

Sound locations

Open **Settings** → **Sound locations**. The catalog tabs are alphabetized: Birdsong, Dolphin Calls, Frog Calls, Ocean Swells, Storm Outlook, and Whale Song. Selecting a tab edits that sound's locations; it does not change the Background choice.

Editable location catalogs

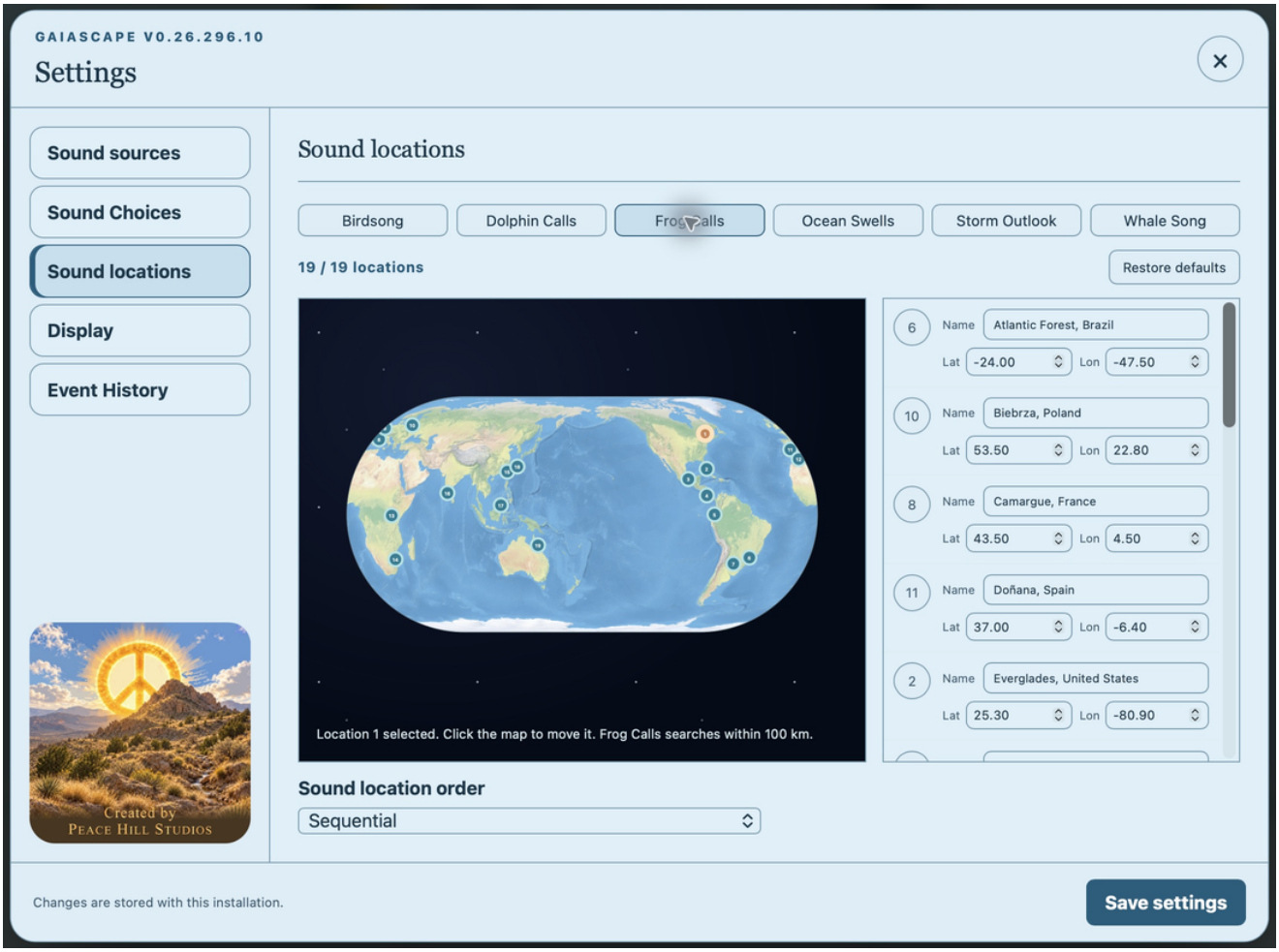


Pacific-centered Ocean Swells location editor with numbered markers, names, coordinates, and Sound location order below the map

Ocean Swells, Storm Outlook, Frog Calls, and Birdsong using Xeno-canto each have **19 locations**. These catalogs are independent.

1. Choose the sound's tab.
2. Select a numbered map marker or the numbered button beside a location in the list. The list is alphabetized by location name; the number links the row to its map marker.
3. Click another position on the map to move the selected location, or edit its **Lat** and **Lon** fields directly.
4. Edit **Name** to give it a meaningful label. Moving a point on the map initially labels it with its new coordinates. Names can be up to 80 characters.
5. Click **Save settings** to apply the edited catalog.

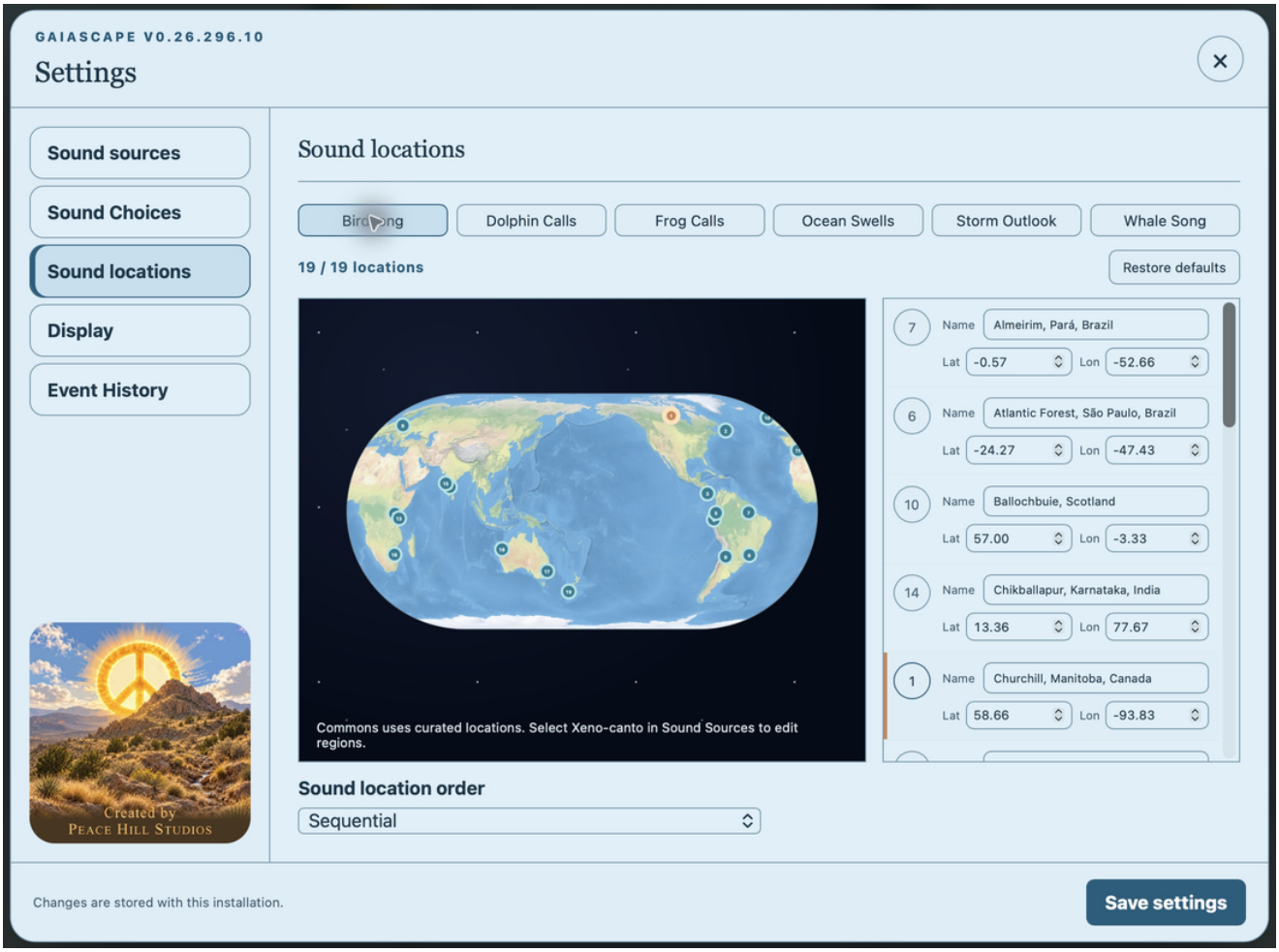
Latitude ranges from **-90 to 90**; longitude ranges from **-180 to 180**. Negative values mean south and west. Keep 19 distinct locations. Ocean Swells points should be at appropriate marine locations; a point on land may not produce usable marine forecasts.



Frog Calls region editor

For **Frog Calls** and **Xeno-canto Birdsong**, each point is a search center with a **100 km radius**. A named region is not a guarantee that suitable recordings exist there. The displayed playback location comes from the actual recording metadata, so it can differ from the center you entered. Frog searches accept short and ungraded recordings with supported CC BY or CC BY-SA licenses; availability still depends on geographic coverage and usable metadata.

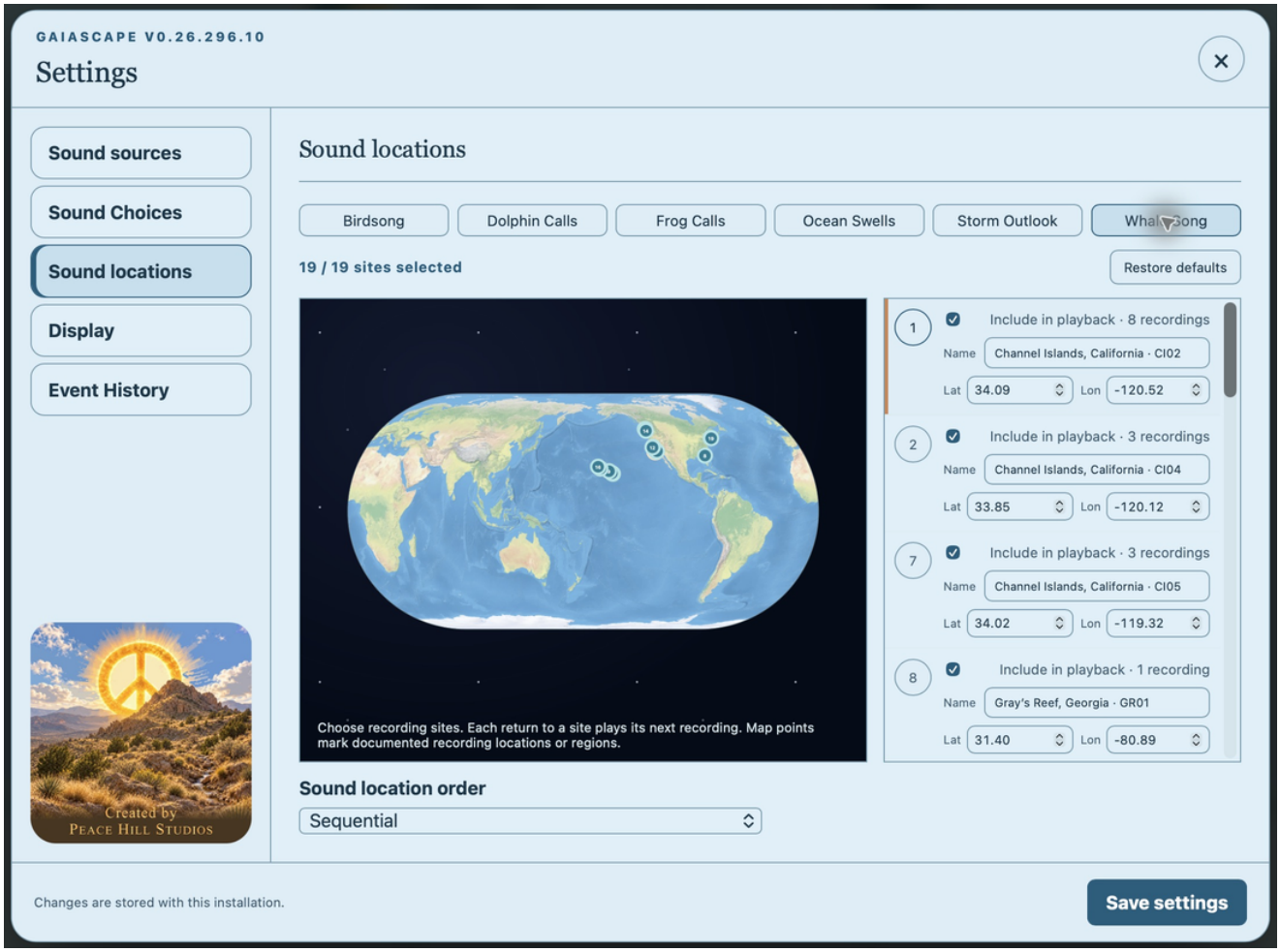
Birdsong using Wikimedia Commons



Birdsong's curated Commons locations with read-only coordinates

Commons uses **68 recordings across 19 curated locations**, with multiple recordings at 17 locations and a single verified recording at two. Each return to a location advances to its next recording. Names and coordinates are read-only, and Restore defaults is disabled for this view. To use editable Birdsong regions, choose **Xeno-canto** in the Birdsong source tile and provide the shared key. Gaiascapes retains the separate Xeno-canto location catalog when you switch providers.

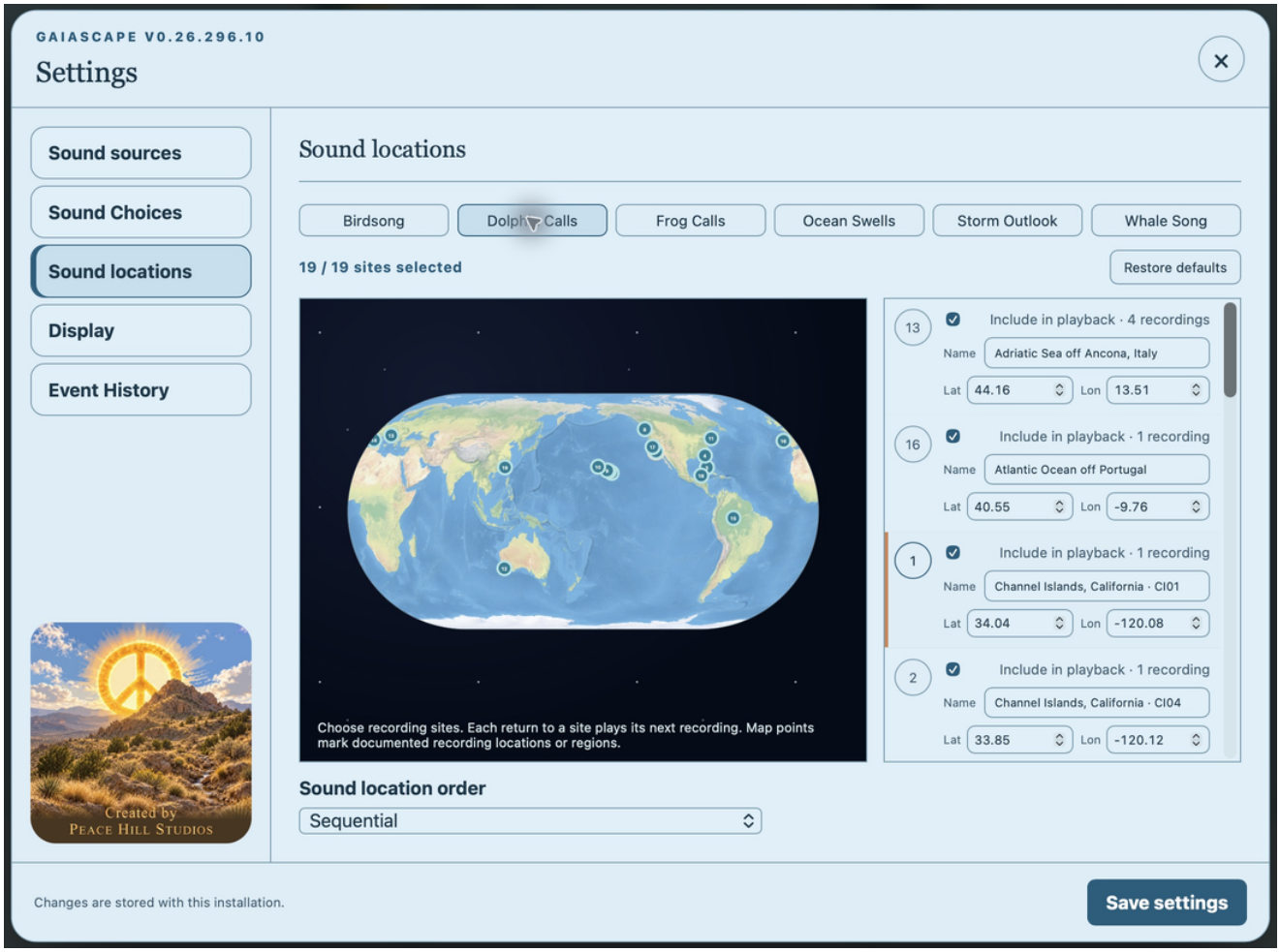
Whale Song and Dolphin Calls



Whale Song's current recording-site selection

These tabs list verified archive sites, not movable search centers. Use **Include in playback** to include or exclude a site from the rotation. Keep at least one site selected. The count shows how many sites are included. The numbered button or map marker identifies the site; the checkbox controls whether it plays. Names and coordinates are read-only.

Catalog	Available sites and regions
Whale Song	**19 sites**, with 59 recordings: California, Hawaiian Islands, Olympic Coast, Gray's Reef, Papahānaumokuākea, and Stellwagen Bank.
Dolphin Calls	**19 sites**, with 30 recordings: the original 11 NOAA sites plus Australia, Italy, Spain, Portugal, Brazil, Mexico, China, and MBARI MARS.



Dolphin Calls' selectable hydrophone sites

Map points identify documented recording locations or approximate study regions, not exact animal positions. Site codes such as HI01 distinguish NOAA recording stations. Each site lists its recording count; single-recording sites repeat on each pass. Playback visits included sites and rotates through available clips at a site. These recordings are archived examples; they are not live hydrophone streams or worldwide searches.

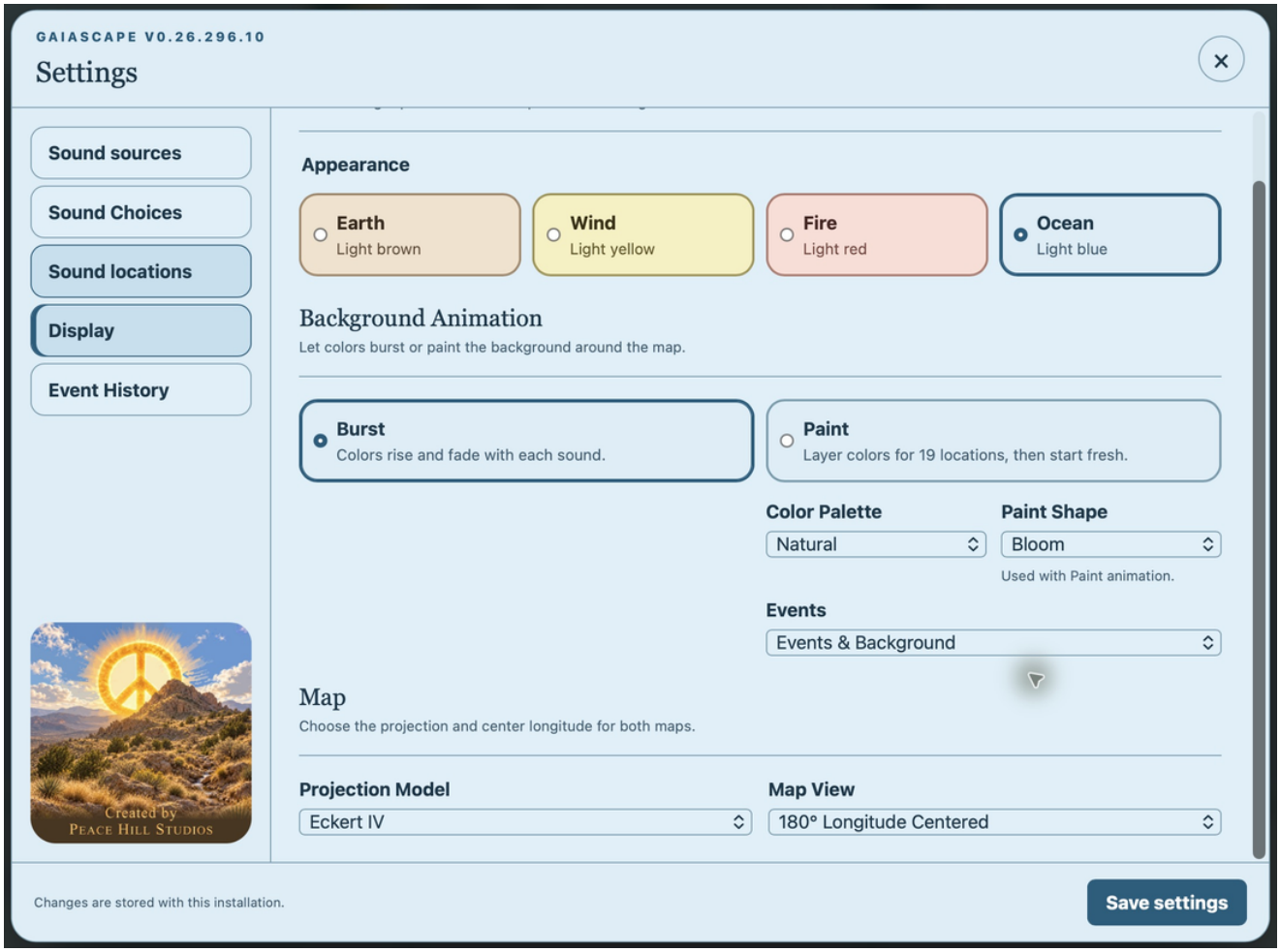
Restore defaults

Restore defaults resets the active catalog in the editor. For editable catalogs, it restores the default 19 points. For Whale Song and Dolphin Calls, it includes every available site again. Click **Save settings** to persist the reset. It does not reset every sound's catalog at once.

Display

Theme

Open **Settings** → **Display** and choose **Earth** (light brown), **Wind** (light yellow), **Fire** (light red), or **Ocean** (light blue). The palette applies to the main view and dialogs immediately, and a **Theme applied.** notification confirms the change.



Display pane with the Theme palettes above Map controls for Projection Model and Map View

Theme previews immediately. Click **Save settings** to store it in this installation's data/config.json; it is restored when the desktop app restarts or a browser reconnects.

Background Animation

Below Theme, choose **Burst** for colors that expand and fade with each background sound, or **Paint** to retain full-size colors around the map's perimeter. Paint layers colors over 19 background location visits, then clears them as the next round begins. Event bursts and map animations keep their usual timing.

Color Palette offers **Natural** (the existing sound colors), **Aurora**, **Wildflower**, and **Pastel**. The additional palettes provide a different shade for each of the 19 location visits. Both selectors sit beneath **Paint** and apply only to Paint animation. **Paint Shape** offers **Bloom** (soft circles), **Watercolor** (irregular washes), and **Ribbon** (curved strokes). **Random** is available in both selectors: each new paint layer independently chooses a palette and/or shape. Existing layers keep their resolved random choices. Changing palette or shape updates existing paint immediately without restarting the round.

Events, beneath the Paint choices, selects **Events & Background** (the default), **Events**, or **Background** for the visible color effects. It filters Paint immediately; Burst continues showing both event and background effects. Audio and map markers continue unchanged. Hidden paint still advances through the 19-location round.

The animation, palette, shape, and Events choices preview immediately. Click **Save settings** to store them with the installation and restore them after restart. Switching modes clears the accumulated paint; reloading starts a fresh canvas.

Map

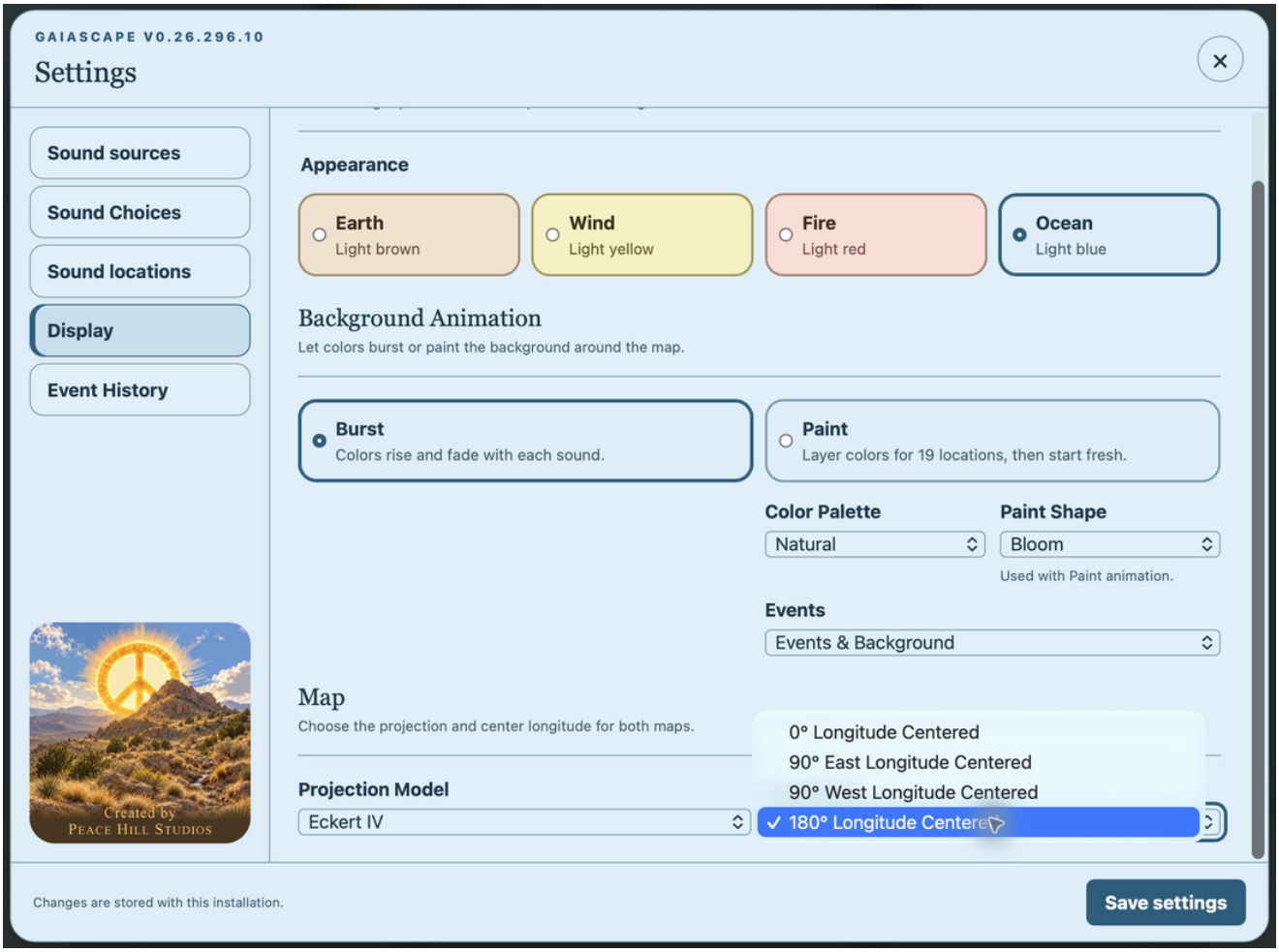
Below Background Animation, **Projection Model** changes both the main map and the location editor:

Projection	Appearance
Robinson	A familiar rounded world map balancing the shapes of continents and oceans.
Eckert IV	An equal-area world map, useful when comparing the relative areas of regions.

Changing projection does not move stored geographic coordinates or change which region supplies sound. Save settings to retain the preference.

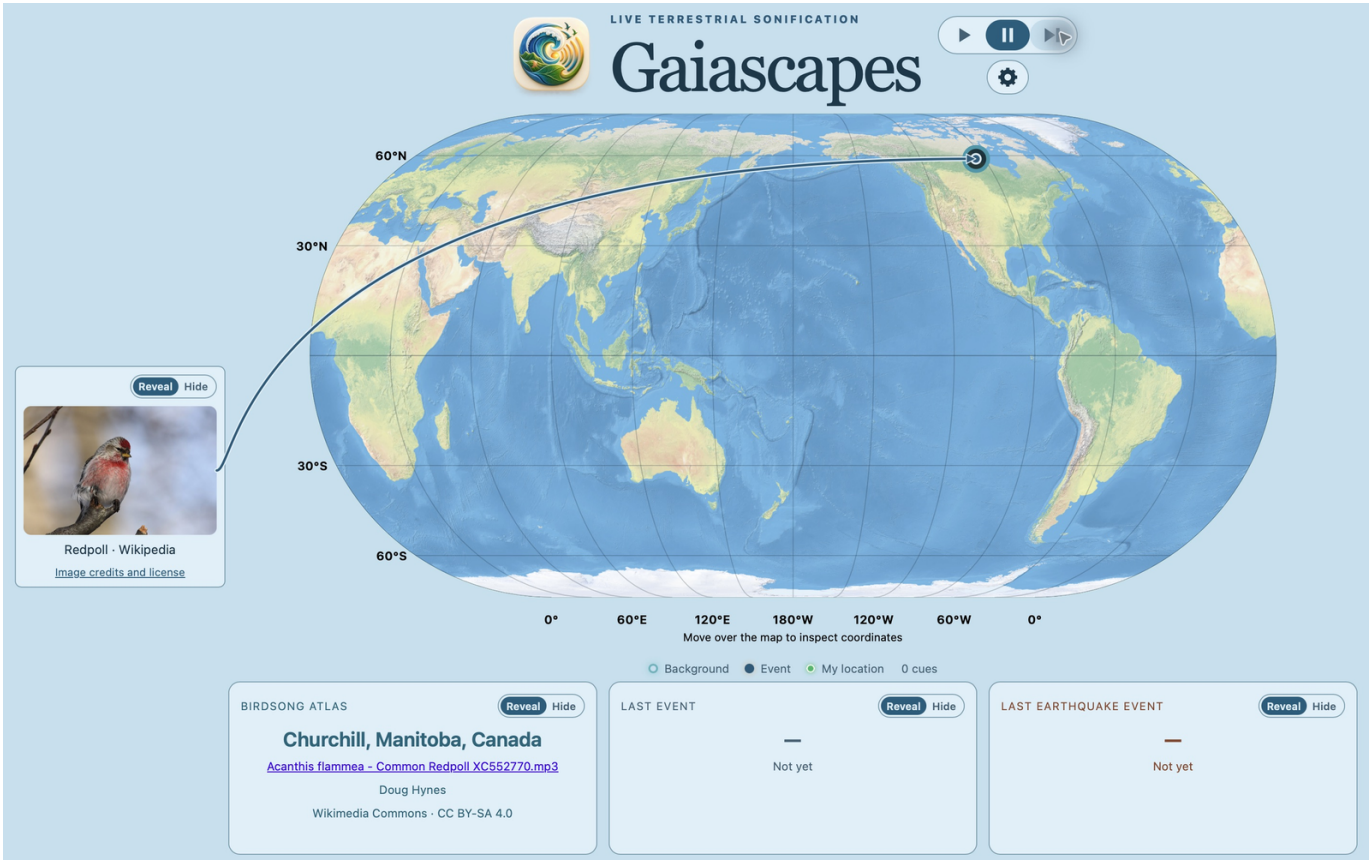
In **Settings** → **Display** → **Map**, choose a **Projection Model** and a **Map View**:

Map View	Center longitude
0° longitude centered	Greenwich meridian; the default.
90° east longitude centered	90° E.
90° west longitude centered	90° W.
180° longitude centered	The antimeridian, centering the Pacific.



Map View expanded to show all four center-longitude choices

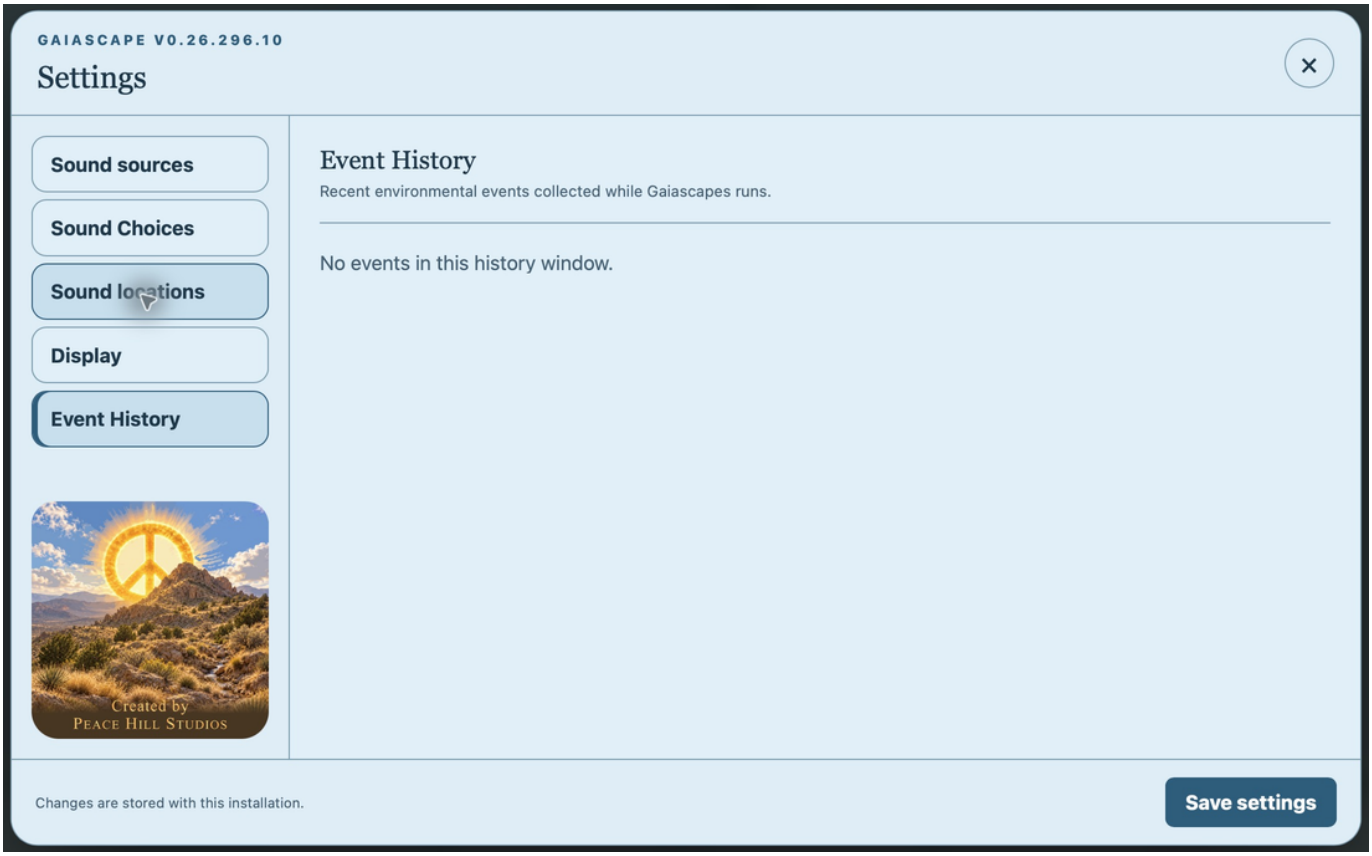
Changing either selector updates both maps immediately. Terrain, grid lines, markers, and coordinate picking follow the selected view; the location catalogs keep their geographic coordinates. Click **Save settings** to remember the map preferences for this installation. Closing without saving leaves the preview visible in the current page; reloading restores the saved settings.



Main map with 180° longitude at its center

Event History

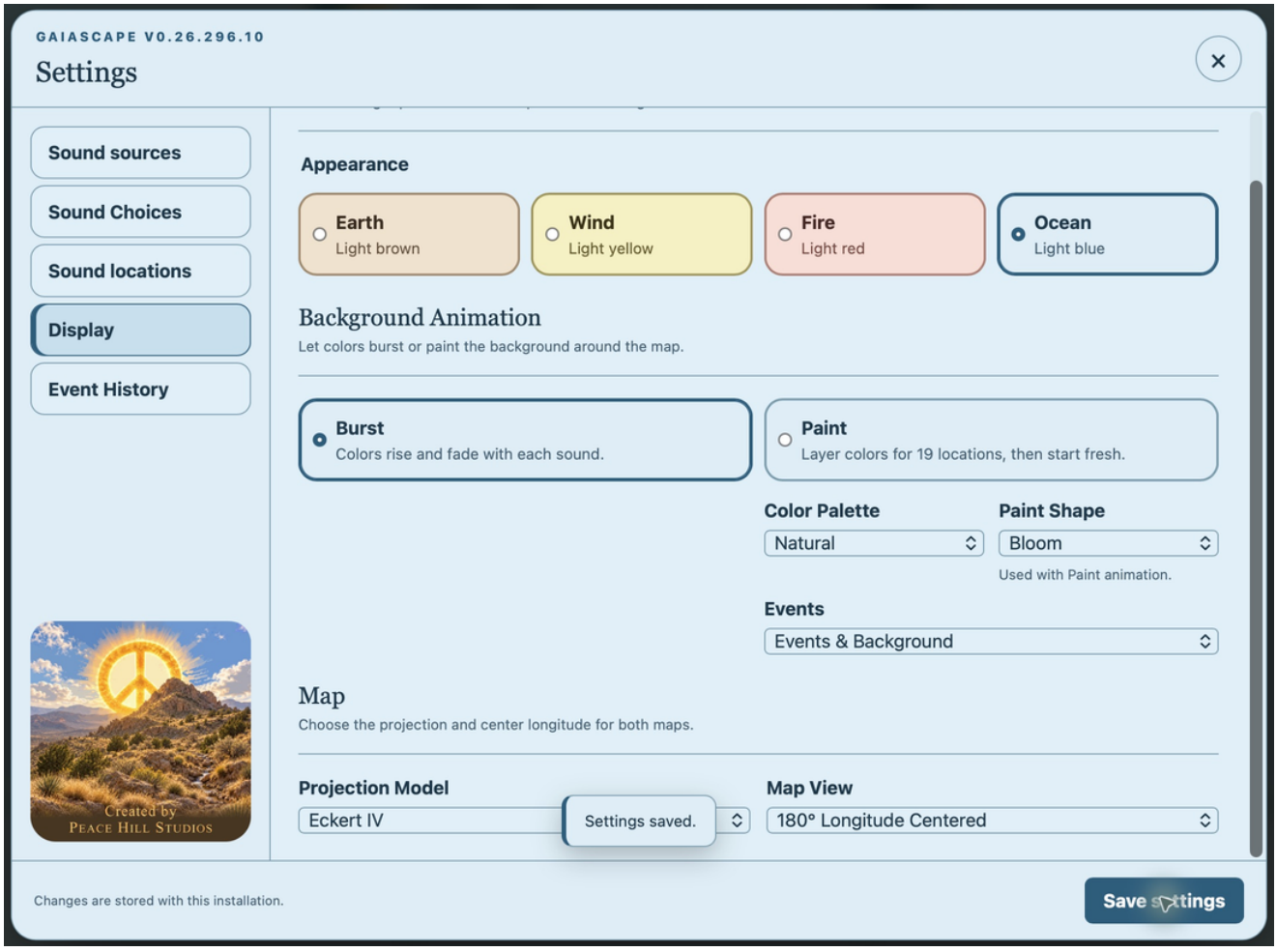
Open **Settings** → **Event History**, the last menu option, to inspect recent activity. Rows show the event type, place, measurements or recording title, associated sound, and time. The list is a presentation of recent activity, not a complete inventory of downloaded recordings. Individual lightning flashes are omitted from the visible history list; they can still trigger sound and map activity. **No events in this history window** means there is no activity to list for the current window.



Event History in a new session with no captured events

Saving settings

Click **Save settings** after editing Sound sources, Sound Choices, Sound locations, or any section in Display. The button saves changes across these sections, not just the visible one. The button shows **Saving…** while the request is in progress. Look for the **Settings saved.** toast before closing the dialog. Success and failure notifications for saving, previewing, restoring defaults, and changing theme dismiss automatically after **five seconds**, or immediately when clicked. Source recovery toasts follow the same behavior. A newer result replaces the previous notification and starts a fresh five-second interval. Errors begin with **Error:**; resolve the reported problem and save again. Earlier parts of a save may already have been applied.



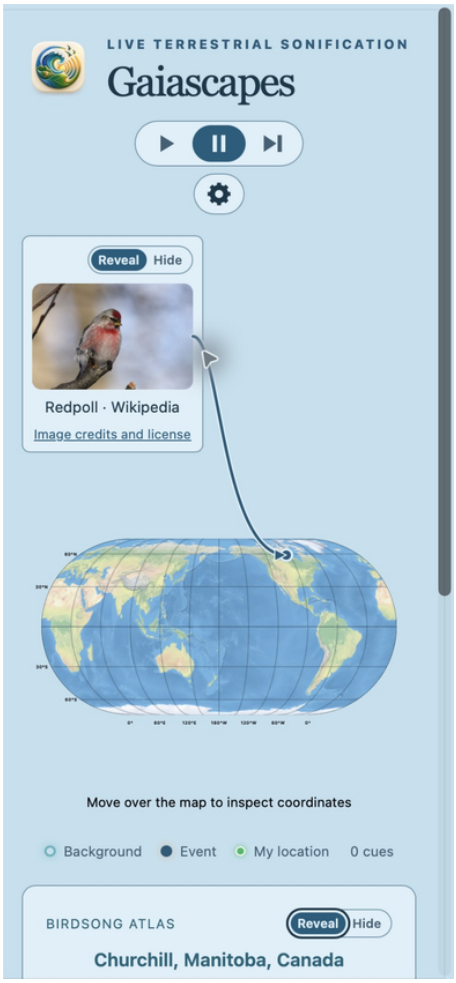
Settings saved notification in Display after saving a Pacific-centered map

Use the × in the dialog header to close Settings; there is no Close button beside Save. Escape or clicking outside the dialog also closes it. Closing is not a save action. Unsaved edits may remain visible when you reopen Settings in the same page, so they should not be treated as a saved configuration. Reloading the page restores the saved settings.

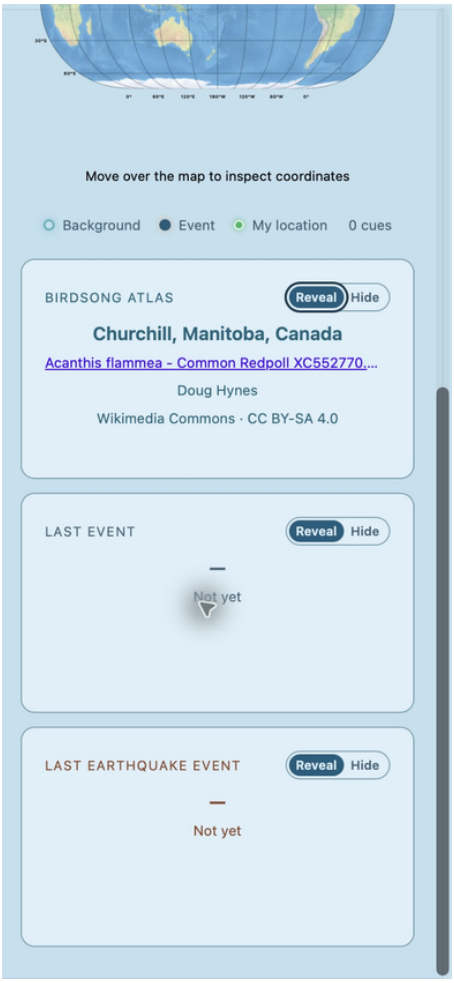
Theme and Background Animation are saved with the installation when you click **Save settings**. Listen/Mute applies only to the current page session. Event History is read-only and requires no save.

Mobile and iPhone Home Screen

On narrow screens, the playback controls and gear remain stacked and centered below the title. The status tiles stack beneath the main content. Scroll down to see all three tiles when they extend below the screen.

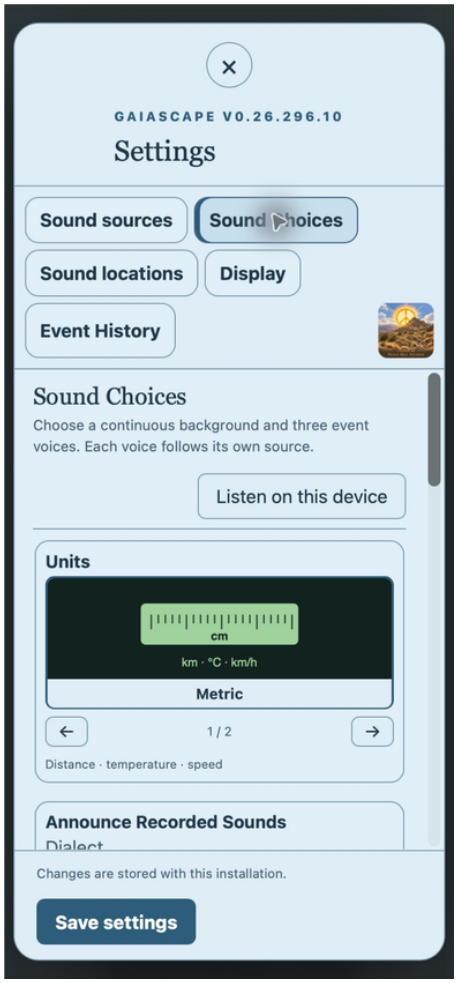


Narrow-screen map with the playback controls, staged animal image, and Pacific-centered map



The same narrow-screen page scrolled down to its three stacked status tiles

In Settings, the section buttons move above the content. The listening button is in the Sound Choices heading. Units, Announce Recorded Sounds, Background, and the event tiles stack below it. Scroll inside the settings content to reach the lower controls. The × remains in the dialog header and **Save settings** in its footer.



Mobile Sound Choices with the listening button in its heading and Units below

To add Gaiascapes to an iPhone Home Screen:

1. Open the running host's LAN URL in **Safari**.
2. Open Safari's **Share** menu and choose **Add to Home Screen**. If it is missing, use **Edit Actions** to add it.
3. Keep **Open as Web App** enabled if Safari offers that option, then tap **Add**. See Apple's Home Screen web-app instructions.
4. Open the new Gaiascapes icon and use **Settings** → **Sound Choices** → **Listen on this device** to enable audio.

Gaiascapes supplies its app artwork for the Home Screen icon. If an existing shortcut still shows a letter, remove that shortcut and add it again from the refreshed site after updating the host. The Home Screen app still connects to the running host; it is not an offline installation. Keeping the phone on the LAN is required, and backgrounding or locking the phone may interrupt audio.

Gaiascapes Game

Turn an animal background into an identification quiz: show the picture first, offer the recording as a second clue, then reveal the answer. One person hosts the game and controls when clues and answers appear.

Set up the quiz

Complete setup before showing the screen to the players.

1. In **Settings** → **Sound sources**, enable the animal recording source you want to use. Clear the checkbox beside **Announce Recording** so spoken names will not give away the answer.
2. In **Sound Choices**, select that animal source as **Background**. Set **Event 1**, **Event 2**, and **Event 3** to **None** to avoid unrelated sounds during the quiz. Set a comfortable Background volume and save settings. If needed, select **Listen on this device** for the browser presenting the quiz.
3. In **Sound locations**, set **Sound location order** to **Random**, then save. Random changes the order of sites within the selected background collection; it does not mix different background sound choices. Mammal sources keep their fixed recording sites.
4. Press **Pause**. The map shows the recording location as a visual clue.
5. Set the background status tile's **Reveal / Hide** selector to **Hide**.
6. Press **Next** to prepare the first animal without playing its sound. Once the picture card appears, set its selector to **Hide** too. The picture and location pointer stay visible, but its caption and credits are concealed.

The screen is now ready for the players. Keep the picture card's Wikipedia article closed until you are ready to reveal the answer.

Play a round

1. **Guess from the picture.** Next displays the animal picture and pauses. Give everyone a chance to identify it without hearing the recording.
2. **Offer the sound if needed.** If nobody knows, press **Play** to hear the same animal. Announcements are disabled, so players hear the animal rather than its name. Press **Pause** after a useful clue; Play resumes continuous playback, which can otherwise advance to another recording automatically.
3. **Reveal the answer.** When someone identifies the animal correctly-or no one can identify it-select **Reveal** on the picture card and background status tile. Read the displayed animal name and recording location together. Some recordings identify only a broad animal group; use the displayed name as the answer rather than requiring an unsupported exact species.
4. **Prepare the next round.** Return both selectors to **Hide**, then press **Next**. The next animal appears silently, with its information still hidden. Repeat the picture, sound, and reveal sequence.

Reveal/Hide choices remain in effect as animals change, but reloading the page resets them to Reveal. If you reload, hide both sets of information again before continuing. To change the animal category, choose another enabled Background in Sound Choices, save, and prepare the next round with Next.

Game controls

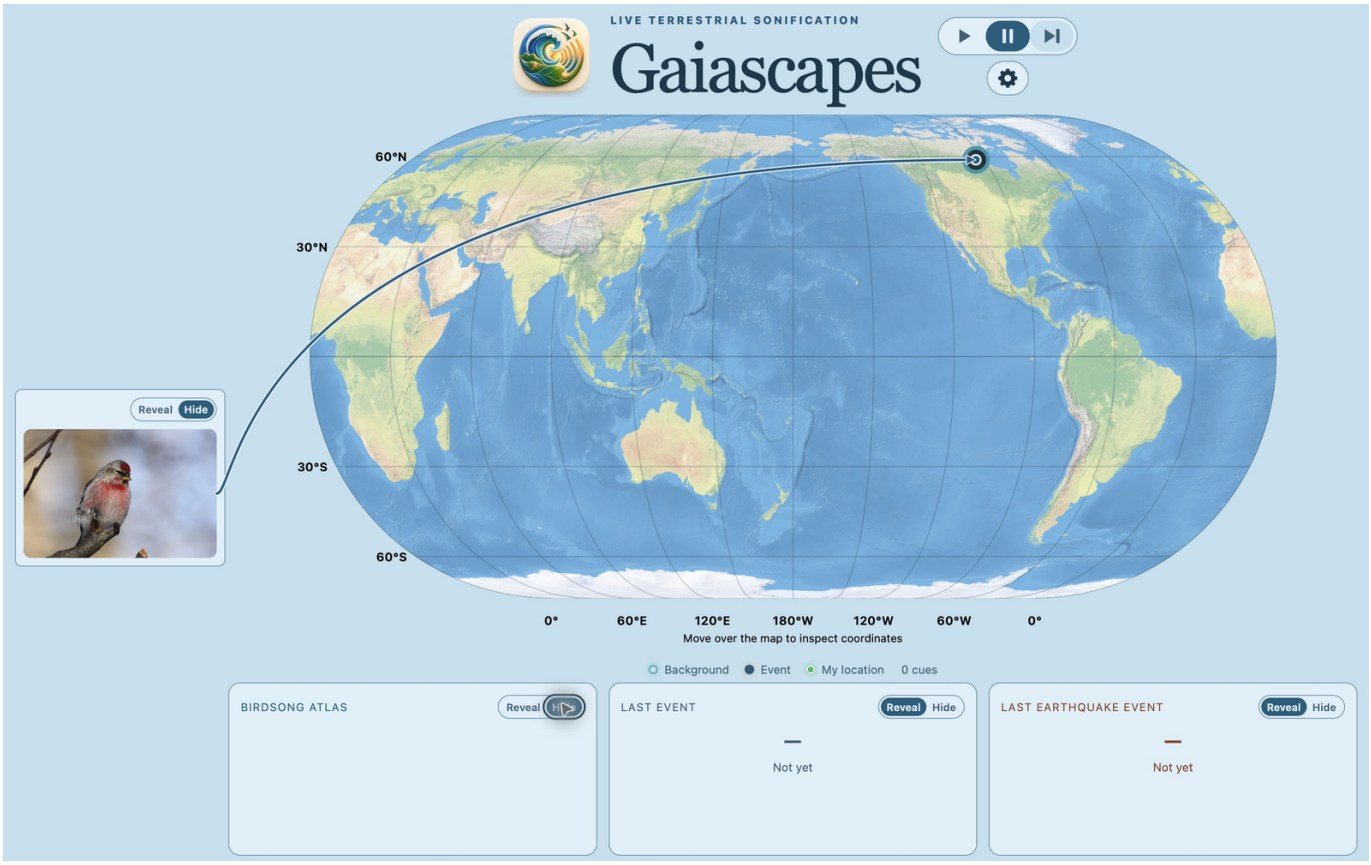
Use the Play, Pause, and Next icons beside the title to control the continuous soundscape. Device listening remains a separate setting.

Each status tile has a **Reveal / Hide** selector beside its title. Reveal is the default. Hide conceals its information, links, and detail tooltip while keeping the title and equal-sized tile visible. Choices apply to the current page; reloading returns to Reveal. New status updates remain hidden until the player selects Reveal.

Sound location order

In **Settings** → **Sound locations**, choose **Sequential** (the default) or **Random** in **Sound location order**, below the map and location list, then save settings. Projection Model is in **Display** → **Map**. Random visits each background location once per shuffled round, then reshuffles; recordings continue to the next clip at each site on subsequent rounds. It applies to birds, frogs, whales, dolphins, the mammal collections, ocean swell, and storm outlook. Actual recording coordinates remain unchanged. Realtime earthquake, thunder/lightning, and tidal events are not randomized. Changes take effect on the next background visit without interrupting a recording.

The animal image tile has its own top-center **Reveal / Hide** selector, initially Reveal. Hide conceals the caption and credits while keeping the image and its map pointer visible; it stays selected as animals change. **Next** stages the next animal in the configured Sequential or Random order and pauses without playing its announcement or recording. Once the image is ready, Pause is highlighted. **Play** plays that staged animal. Next requires an enabled animal recording background.



Quiz view with the Redpoll image and location pointer visible while its caption, credits, and background status details are hidden

Status and troubleshooting

What you see or hear	What it means and what to check
Remote page will not open	Verify the host is running, use its LAN IP with port 8769, and check firewall and Wi-Fi client isolation.
Listening stops after locking the phone or switching apps	Return to Gaiascapes and follow the message beside Listen to reconnect. Browser audio can be suspended in the background.
No background selected	Background is set to None.
Awaiting background	No matching background cue is available yet. This alone does not identify a provider failure. Check that playback is running and the source is enabled.
Looking for … recordings…	A recording lookup or download is underway. First-use requests can take longer than cached playback.
No suitable recordings … within 100 km …	The lookup completed without a recording that meets the region and compatibility requirements. Continuous playback advances to the next region on a later cycle. Adjust a search center if needed; the app does not substitute another animal group or a recording outside the region.
Rejected key, denied access, or rate limit	Check the credential instructions above. Respect a displayed retry delay. Replacing the shared Xeno-canto key affects both Birdsong and Frog Calls.
Preview reports a provider error	Source selection, credentials, geography, or download availability prevented a preview. The error should explain the reason.
Recording is ready, but silent	Check the Background volume, media volume, and browser/site audio permissions. On a remote device, tap Listen on this device to permit playback. Gaiascapes requests a media playback session on supported browsers so iPhone Silent Mode does not mute explicit listening. With an older browser or app version, try turning Silent Mode off, then reconnect.
Recordings work but synthesized sounds are silent	For LAN listening, restart the updated SuperCollider receiver on the Gaiascapes host and tap Listen again. Check the configured renderer and audio output for host playback.
Event voices are quiet while the background plays	Events are intermittent. Check the event slots, their volumes, and their corresponding source switches. Thunder requires observed lightning; Tidal Tone requires a modeled tide turn.
A saved sound has no effect	Confirm the save succeeded and inspect the running version. The app can only show features present in that running release.

Automatic source-recovery notices can appear when an environmental feed fails or recovers. A source problem does not necessarily stop other enabled sources. A cached recording can remain playable even while new network lookups are unavailable.

Location privacy

In **Settings** → **Sound Choices**, clear **Show this host's approximate location on the map** and save to stop IP-based location lookups. This hides the host marker without changing environmental events, audio, or LAN access. The marker refers to the Gaiascapes host, not the phone or other listening device. When enabled, IP location also sets the initial location 1 for Xeno-canto Birdsong and Storm Outlook at startup; Frog Calls retains its verified defaults or edited regions. The setting applies to the installation; other open browsers may need a refresh to remove a previously displayed marker. See Privacy for details.

Screenshot image credit

The Redpoll photograph in the refreshed map, mobile, and quiz screenshots is Carduelis flammea CT6 by Cephas, used under CC BY-SA 3.0. It is scaled and cropped by the application's image card. The photograph retains its license. The photograph's location is separate from the recording location shown on the map; the screenshot uses the Commons recording selected by Next.